Introduction

The objects in our everyday lives have an annoying inability to appear in two places at once. Leave your keys in your jacket, and they won't also be on the hook by the front door. This isn't surprising—these objects have no uncharted abilities or virtues. They're profoundly ordinary. Yet these mundane things are composed of a galaxy of the unfamiliar. Your house keys are a temporary alliance of a trillion trillion atoms, each forged in a dying star eons ago, each falling to Earth in its earliest days. They have bathed in the light of a violent young sun. They have witnessed the entire history of life on our planet. Atoms are epic.

Like most epic heroes, atoms have some problems that ordinary humans don't. We are creatures of habit, monotonously persisting in just one location at a time. But atoms are prone to whimsy. A single atom, wandering down a path in a laboratory, encounters a fork where it can go left or right. Rather than choosing one way forward, as you or I would have to do, the atom suffers a crisis of indecision over where to be and where not to be. Ultimately, our nanometer Hamlet chooses both. The atom doesn't split, it doesn't take one path and then the other—it travels down both paths, simultaneously, thumbing its nose at the laws of logic. The rules that apply to you and me and Danish princes don't apply to atoms. They live in a different world, governed by a different physics: the submicroscopic world of the quantum.

Quantum physics—the physics of atoms and other ultratiny objects, like molecules and subatomic particles—is the most successful theory in all of science. It predicts a stunning variety of phenomena to an extraordinary degree of accuracy, and its impact goes well beyond the world of the very small and into our everyday lives. The discovery of quantum physics in the